

[illegible]

CPU Address Bus Buffering

This section shows the buffering of CPU address pins (CPU_A0_Pin to CPU_A13_Pin) to internal CPU address signals (CPU_A0{H4} to CPU_A12{H27}). It uses inverters and buffers to drive the CPU address bus.

M2 Clock Buffering

This section shows the buffering of the M2 clock signal (M2_Pin) to internal M2 clock signals (M2_CLK_BUF1{BM21} to M2_CLK_BUF3{A38}). It uses buffers and inverters to drive the M2 clock bus.

CPU Address Bus Nybble Decoding

This section shows the decoding of CPU address nybbles (CPU_A7{H18} to CPU_A4{H12}) to internal CPU address signals (CPU_2xxx{J53} to CPU_5xxx{K84}). It uses NAND gates and inverters to drive the CPU address bus.

CPU R/W, Register Read/Write

This section shows the decoding of CPU R/W signals (CPU_R/W_Pin) to internal CPU R/W signals (Reg_Read{K5} to Reg_Write{L111}). It uses buffers and inverters to drive the CPU R/W bus.

/ROMSEL, Register Access

This section shows the decoding of the /ROMSEL signal (/ROMSEL_Pin) to internal /ROMSEL signals (Reg_Access{H36} to Reg_Access{H2}). It uses buffers and inverters to drive the /ROMSEL bus.

Notes:

- The high nybbles are not taking /ROMSEL or A15 into account, so they are 32kB mirrored at this point.
- "Reg_Access" is included in downstream logic to specify which 32kB region.
- Issues seeing traces in this area.

**Sheet: /CPU Address Bus/
File: cpuAddressBus.kicad_sch**

Title:

Size: USLdger **Date:** **Rev:**

KiCad E.D.A. 9.0.7 **Id:** 2/26

This KICAD schematic illustrates the CPU Address Bus Buffering and Nybble Decoding logic. The design is organized into several functional blocks:

- CPU Address Bus Buffering:** This block shows the buffering of CPU address pins (CPU_A0_Pin to CPU_A13_Pin) and the M2 clock signal (M2_Pin). The CPU address pins are buffered and then decoded into various CPU address signals (e.g., /CPU_A0{H4}, +CPU_A0{H3}). The M2 clock signal is buffered and then decoded into various M2 clock signals (e.g., /M2_CLK_BUF1{BM21}, +M2_CLK_BUF1{BM21}).
- M2 Clock Buffering:** This block shows the buffering of the M2 clock signal (M2_Pin) and the M2 clock buffer signals (M2_CLK_BUF1, M2_CLK_BUF2, M2_CLK_BUF3). The M2 clock signal is buffered and then decoded into various M2 clock signals (e.g., /M2_CLK_BUF1{BM21}, +M2_CLK_BUF1{BM21}).
- CPU Address Bus Nybble Decoding:** This block shows the decoding of CPU address signals into various CPU address nybbles (e.g., /CPU_0xx0x{J41}, +CPU_0xx0x{J42}). The CPU address signals are decoded into various CPU address nybbles (e.g., /CPU_0xx0x{J41}, +CPU_0xx0x{J42}).
- /ROMSEL, Register Access:** This block shows the decoding of the /ROMSEL signal into various register access signals (e.g., /Reg_Read{K5}, +Reg_Read{K3}). The /ROMSEL signal is decoded into various register access signals (e.g., /Reg_Read{K5}, +Reg_Read{K3}).
- CPU R/W, Register Read/Write:** This block shows the decoding of the CPU R/W signal into various register read/write signals (e.g., /Reg_Read{K5}, +Reg_Read{K3}). The CPU R/W signal is decoded into various register read/write signals (e.g., /Reg_Read{K5}, +Reg_Read{K3}).

The schematic uses a variety of logic components, including buffers, inverters, NAND gates, and decoders, to implement the address bus buffering and nybble decoding logic. The components are labeled with their respective part numbers and pin numbers.

Sheet: /CPU Address Bus/
File: cpuAddressBus.kicad_sch

Title:

Size: USLdger Date: Rev:
KiCad E.D.A. 9.0.7 Id: 2/26

CPU Address Bus Buffering

M2 Clock Buffering

CPU Address Bus Nybble Decoding

CPU R/W, Register Read/Write

/ROMSEL, Register Access

Note: The high nybbles are not taking /ROMSEL or A15 into account, so they are 32kB mirrored at this point. "Reg_Access" is included in downstream logic to specify which 32kB region.

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CPU Address Bus Buffering

M2 Clock Buffering

CPU Address Bus Nybble Decoding

CPU R/W, Register Read/Write

/ROMSEL, Register Access

Note: The high nybbles are not taking /ROMSEL or A15 into account, so they are 32kB mirrored at this point. "Reg_Access" is included in downstream logic to specify which 32kB region.

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CPU Address Bus Buffering

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CPU Address Bus Nybble Decoding

CPU R/W, Register Read/Write

/ROMSEL, Register Access

Note: The high nybbles are not taking /ROMSEL or A15 into account, so they are 32kB mirrored at this point. "Reg_Access" is included in downstream logic to specify which 32kB region.

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CPU Address Bus Buffering

M2 Clock Buffering

CPU Address Bus Nybble Decoding

CPU R/W, Register Read/Write

/ROMSEL, Register Access

Note: The high nybbles are not taking /ROMSEL or A15 into account, so they are 32kB mirrored at this point. "Reg_Access" is included in downstream logic to specify which 32kB region.

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CPU Address Bus Buffering

This section shows the buffering of CPU address pins (CPU_A0_Pin to CPU_A13_Pin) to internal CPU address signals (CPU_A0{H4} to CPU_A12{H27}). It uses inverters and buffers to drive the CPU address bus.

M2 Clock Buffering

This section shows the buffering of the M2 clock signal (M2_Pin) to internal M2 clock signals (M2_CLK_BUF1{BM21} to M2_CLK_BUF3{A38}). It uses buffers and inverters to drive the M2 clock bus.

CPU Address Bus Nybble Decoding

This section shows the decoding of CPU address nybbles (CPU_A7{H18} to CPU_A4{H12}) to internal CPU address signals (CPU_2xxx{J53} to CPU_5xxx{K84}). It uses NAND gates and inverters to drive the CPU address bus.

CPU R/W, Register Read/Write

This section shows the decoding of CPU R/W signals (CPU_R/W_Pin) to internal CPU R/W signals (Reg_Read{K5} to Reg_Write{L111}). It uses buffers and inverters to drive the CPU R/W bus.

/ROMSEL, Register Access

This section shows the decoding of the /ROMSEL signal (/ROMSEL_Pin) to internal /ROMSEL signals (Reg_Access{H36} to Reg_Access{H2}). It uses buffers and inverters to drive the /ROMSEL bus.

Notes:

- The high nybbles are not taking /ROMSEL or A15 into account, so they are 32kB mirrored at this point.
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- Issues seeing traces in this area.

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File: cpuAddressBus.kicad_sch
Title:
Size: USLdger **Date:** **Rev:**
KiCad E.D.A. 9.0.7 **Id: 2/26**

CPU Address Bus Buffering

This section shows the buffering of CPU address pins (CPU_A0_Pin to CPU_A13_Pin) through inverters and buffers to generate internal CPU address signals (e.g., /CPU_A0{H4}, +CPU_A0{H3}).

M2 Clock Buffering

This section shows the buffering of the M2 clock signal (M2_Pin) through buffers and inverters to generate internal clock signals (e.g., /M2_CLK_BUF1{BM21}, +M2_CLK_BUF1{BM21}).

CPU Address Bus Nybble Decoding

This section shows the decoding of CPU address nybbles (e.g., /CPU_A7{H18}, /CPU_A6{H16}) using NAND gates and inverters to generate internal signals (e.g., /CPU_2xxx{J53}, +CPU_2xxx{J54}).

CPU R/W, Register Read/Write

This section shows the logic for CPU Read/Write signals (CPU_R/W_Pin) and register access signals (e.g., /Reg_Read{K5}, +Reg_Read{K3}).

/ROMSEL, Register Access

This section shows the logic for the /ROMSEL signal (E23, E24) and register access signals (e.g., +Reg_Access{H36}, /Reg_Access{H37}).

Pinout Table

Pin	Signal	Pin	Signal	Pin	Signal	Pin	Signal
1	CPU_A0_Pin	17	CPU_A10_Pin	33	CPU_A14_Pin	49	CPU_A11{H26}
2	CPU_A1_Pin	18	CPU_A11_Pin	34	CPU_A15_Pin	50	CPU_A12{H27}
3	CPU_A2_Pin	19	CPU_A12_Pin	35	CPU_A16_Pin	51	CPU_A13{H30}
4	CPU_A3_Pin	20	CPU_A13_Pin	36	CPU_A17_Pin	52	CPU_A14{H32}
5	CPU_A4_Pin	21	CPU_A14_Pin	37	CPU_A18_Pin	53	CPU_A15{H33}
6	CPU_A5_Pin	22	CPU_A15_Pin	38	CPU_A19_Pin	54	CPU_A16{H34}
7	CPU_A6_Pin	23	CPU_A16_Pin	39	CPU_A20_Pin	55	CPU_A17{H35}
8	CPU_A7_Pin	24	CPU_A17_Pin	40	CPU_A21_Pin	56	CPU_A18{H36}
9	CPU_A8_Pin	25	CPU_A18_Pin	41	CPU_A22_Pin	57	CPU_A19{H37}
10	CPU_A9_Pin	26	CPU_A19_Pin	42	CPU_A23_Pin	58	CPU_A20{H38}
11	CPU_A10_Pin	27	CPU_A20_Pin	43	CPU_A24_Pin	59	CPU_A21{H39}
12	CPU_A11_Pin	28	CPU_A21_Pin	44	CPU_A25_Pin	60	CPU_A22{H40}
13	CPU_A12_Pin	29	CPU_A22_Pin	45	CPU_A26_Pin	61	CPU_A23{H41}
14	CPU_A13_Pin	30	CPU_A23_Pin	46	CPU_A27_Pin	62	CPU_A24{H42}
15	CPU_A14_Pin	31	CPU_A24_Pin	47	CPU_A28_Pin	63	CPU_A25{H43}
16	CPU_A15_Pin	32	CPU_A25_Pin	48	CPU_A29_Pin	64	CPU_A26{H44}

CPU Address Bus Buffering

M2 Clock Buffering

CPU Address Bus Nybble Decoding

CPU R/W, Register Read/Write

/ROMSEL, Register Access

Note: The high nybbles are not taking /ROMSEL or A15 into account, so they are 32kB mirrored at this point. "Reg_Access" is included in downstream logic to specify which 32kB region.

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KiCad E.D.A. 9.0.7 Id: 2/26

CPU Address Bus Buffering

M2 Clock Buffering

CPU Address Bus Nybble Decoding

CPU R/W, Register Read/Write

/ROMSEL, Register Access

Note: The high nybbles are not taking /ROMSEL or A15 into account, so they are 32kB mirrored at this point. "Reg_Access" is included in downstream logic to specify which 32kB region.

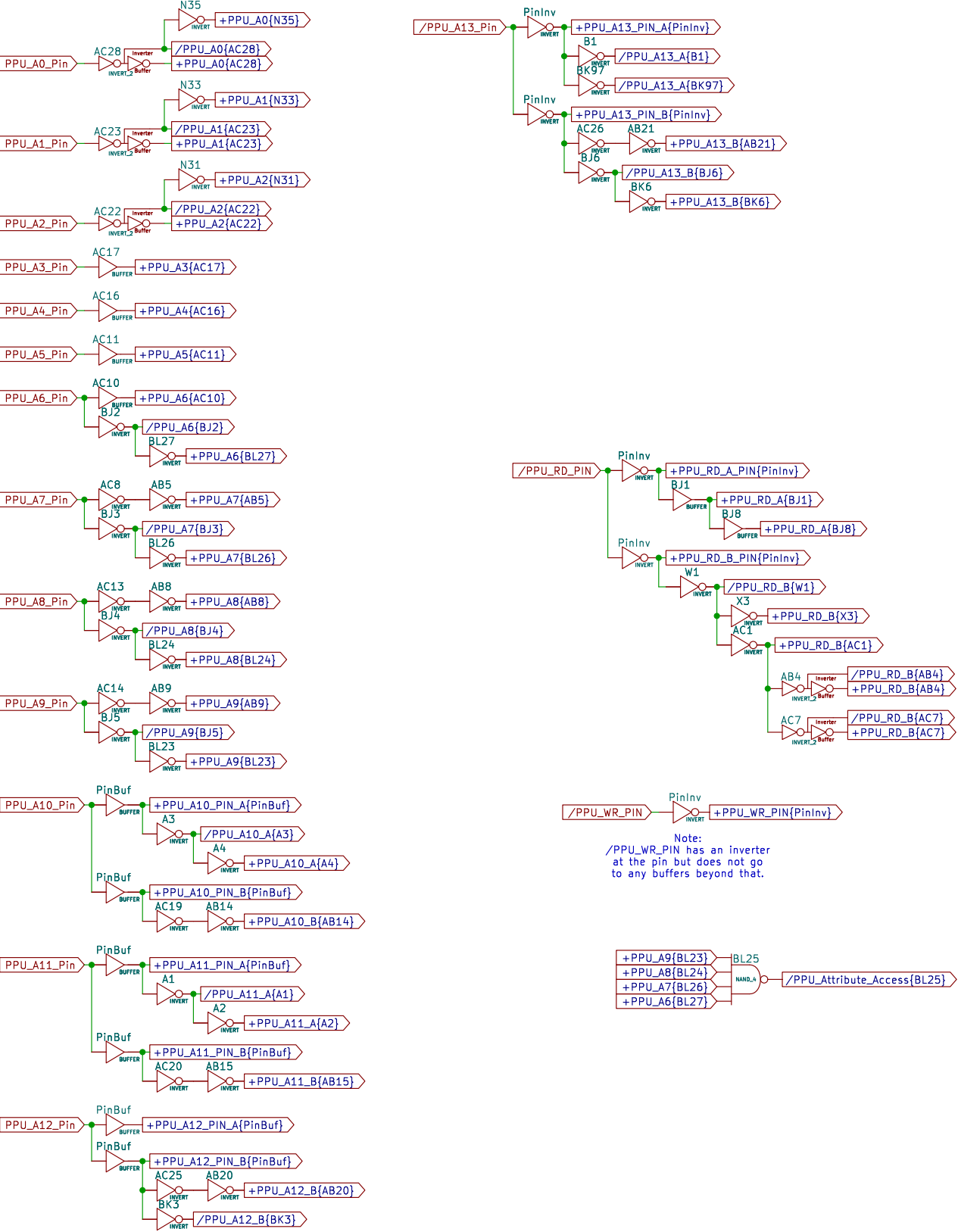
Issues seeing traces in this area.

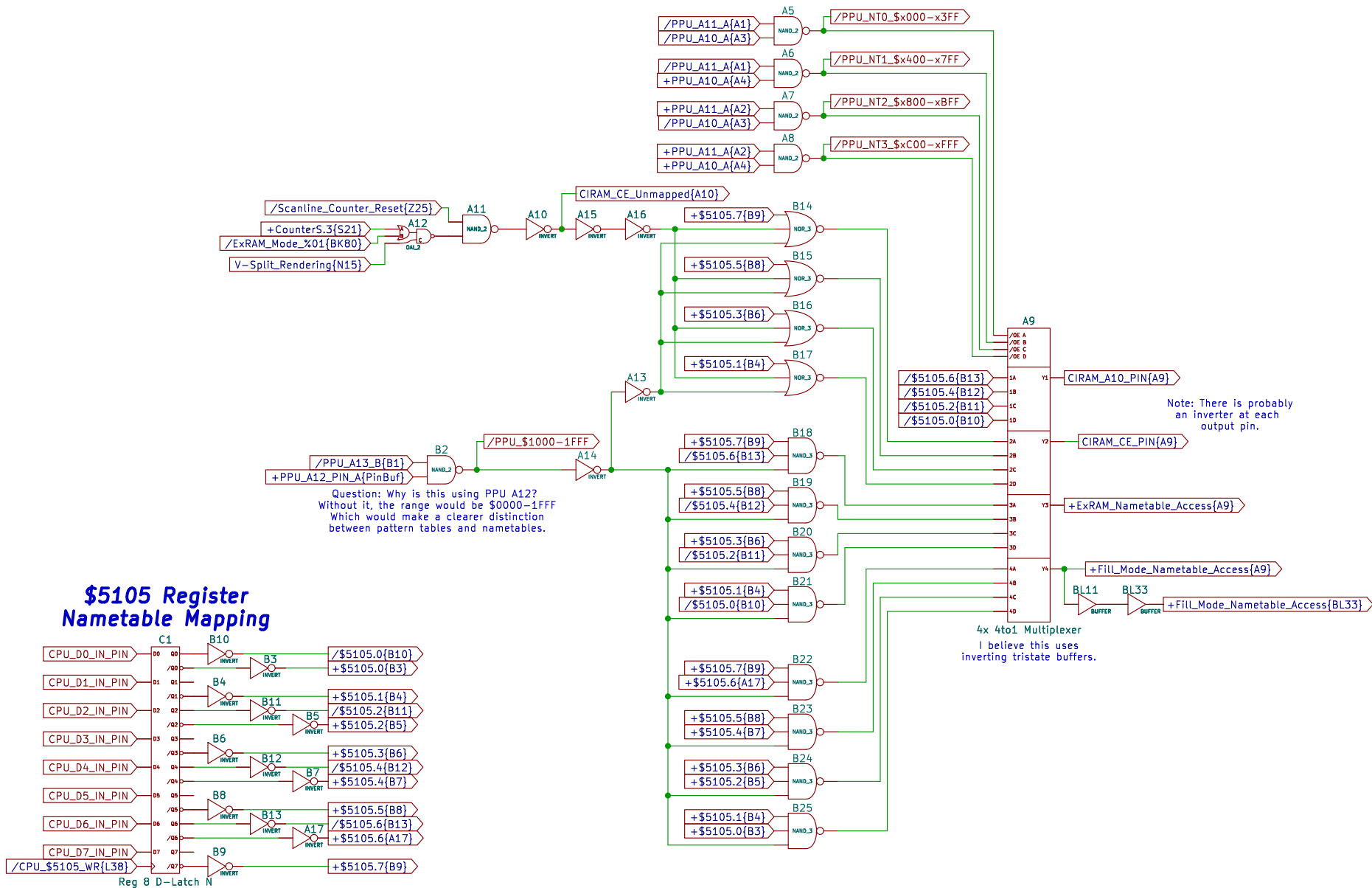
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File: cpuAddressBus.kicad_sch

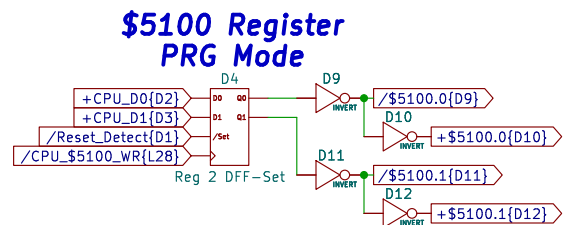
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Size: USLdger Date: Rev:
KiCad E.D.A. 9.0.7 Id: 2/26

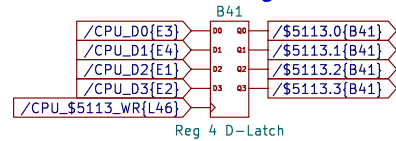
PPU Address Bus Buffering



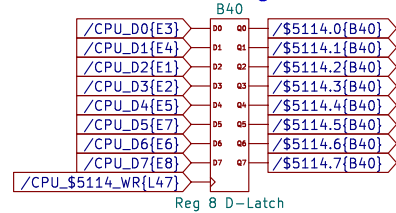




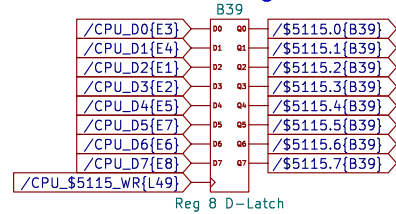
\$5113 Register



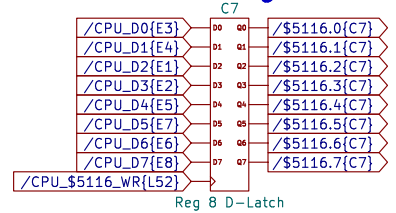
\$5114 Register



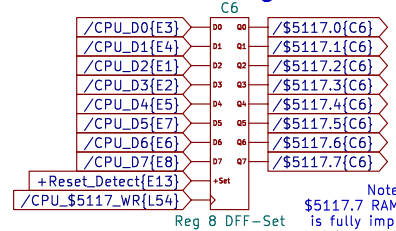
\$5115 Register



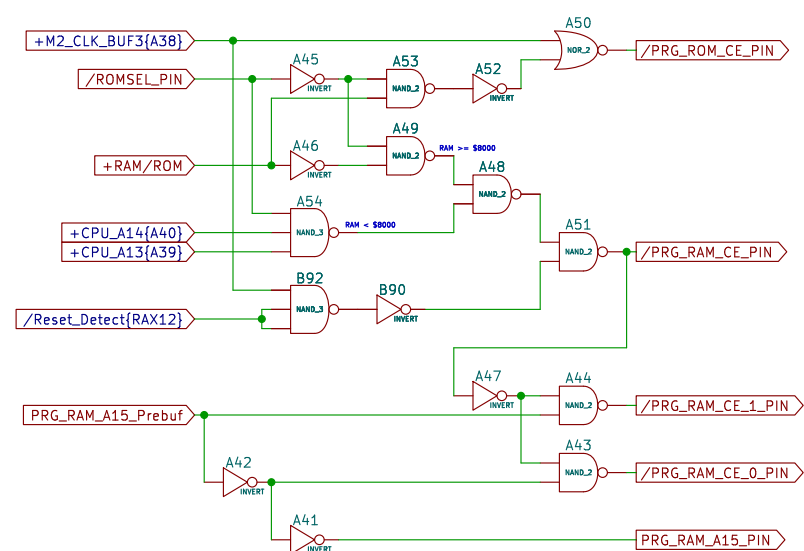
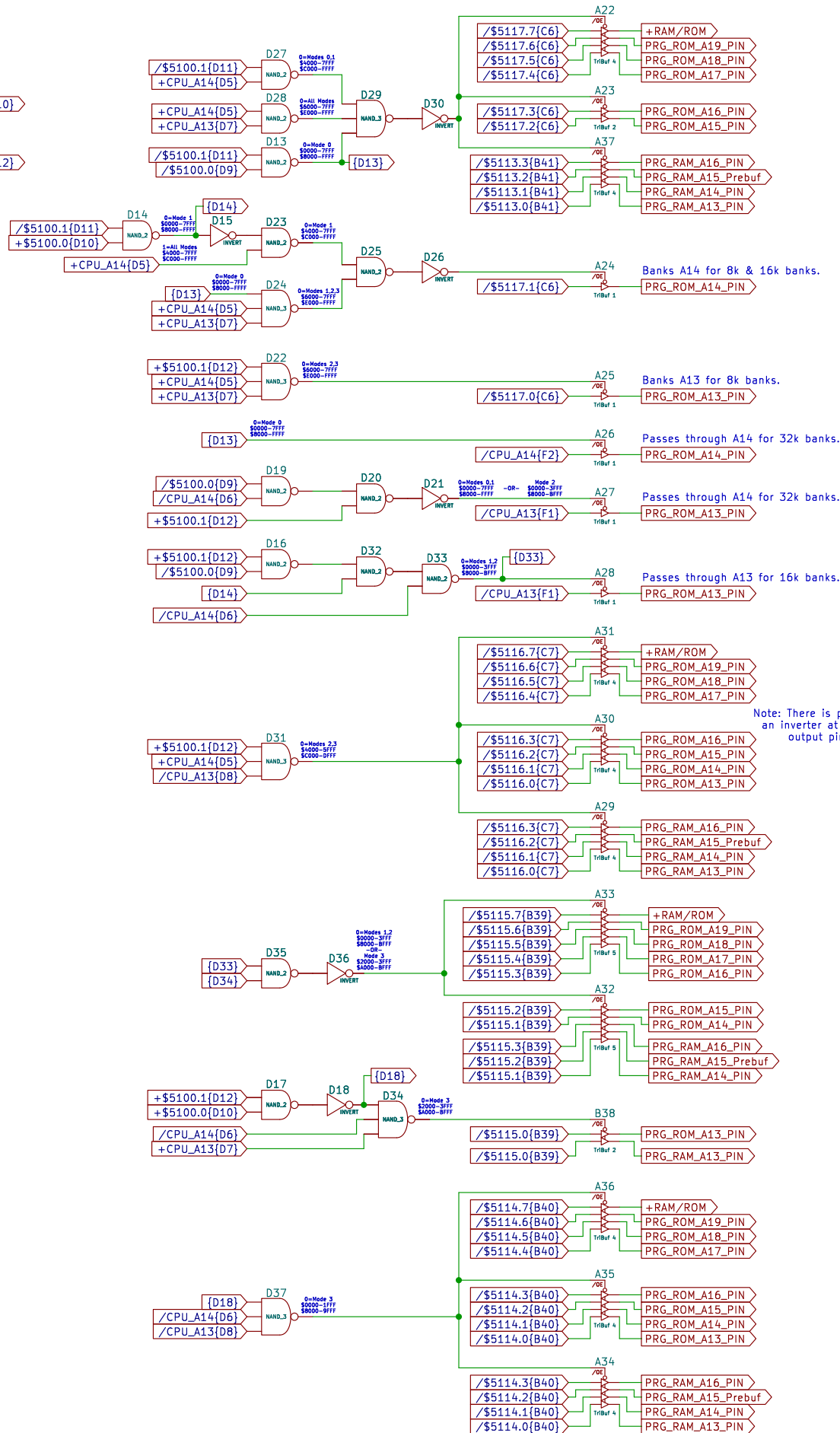
\$5116 Register



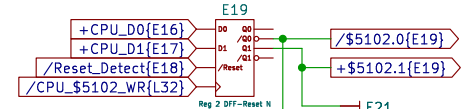
\$5117 Register



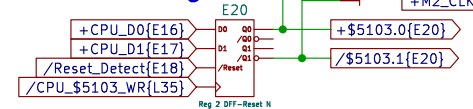
Note:
\$5117.7 RAM/ROM bit
is fully implemented
but is somehow being
prevented from entering
RAM mode when bench
testing.



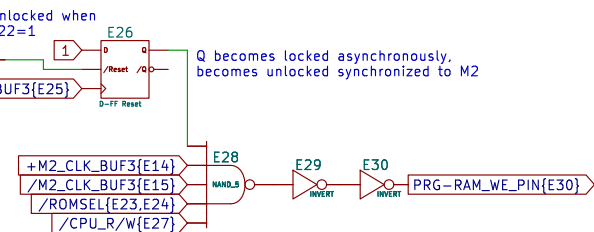
\$5102 Register PRG-RAM Write Protection

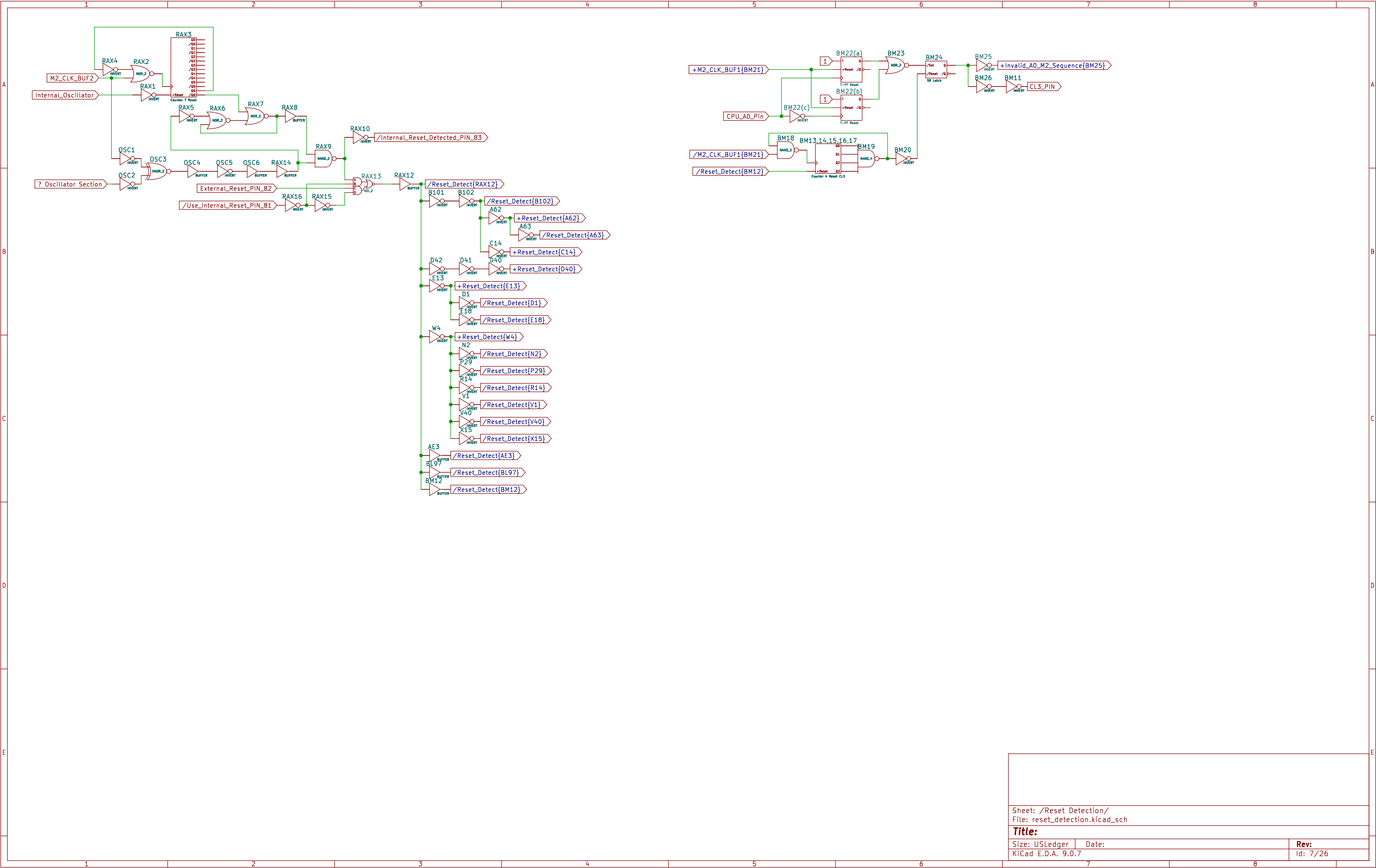


\$5103 Register



Note: /Set vs. /Reset inputs are individually used but not easily shown here with a symbol.





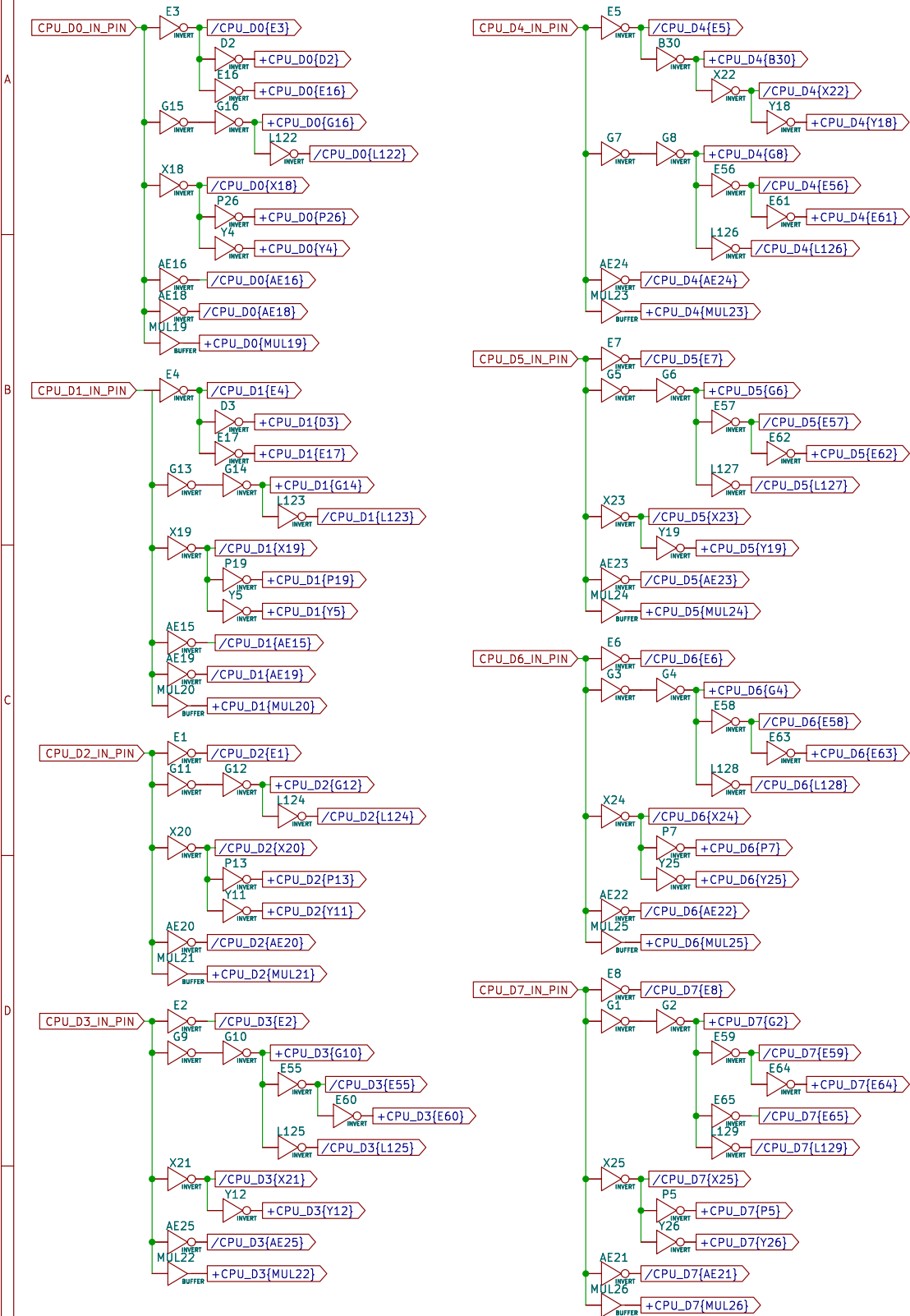
Sheet: /Reset Detection/
File: reset_detection.kicad_sch

Title:

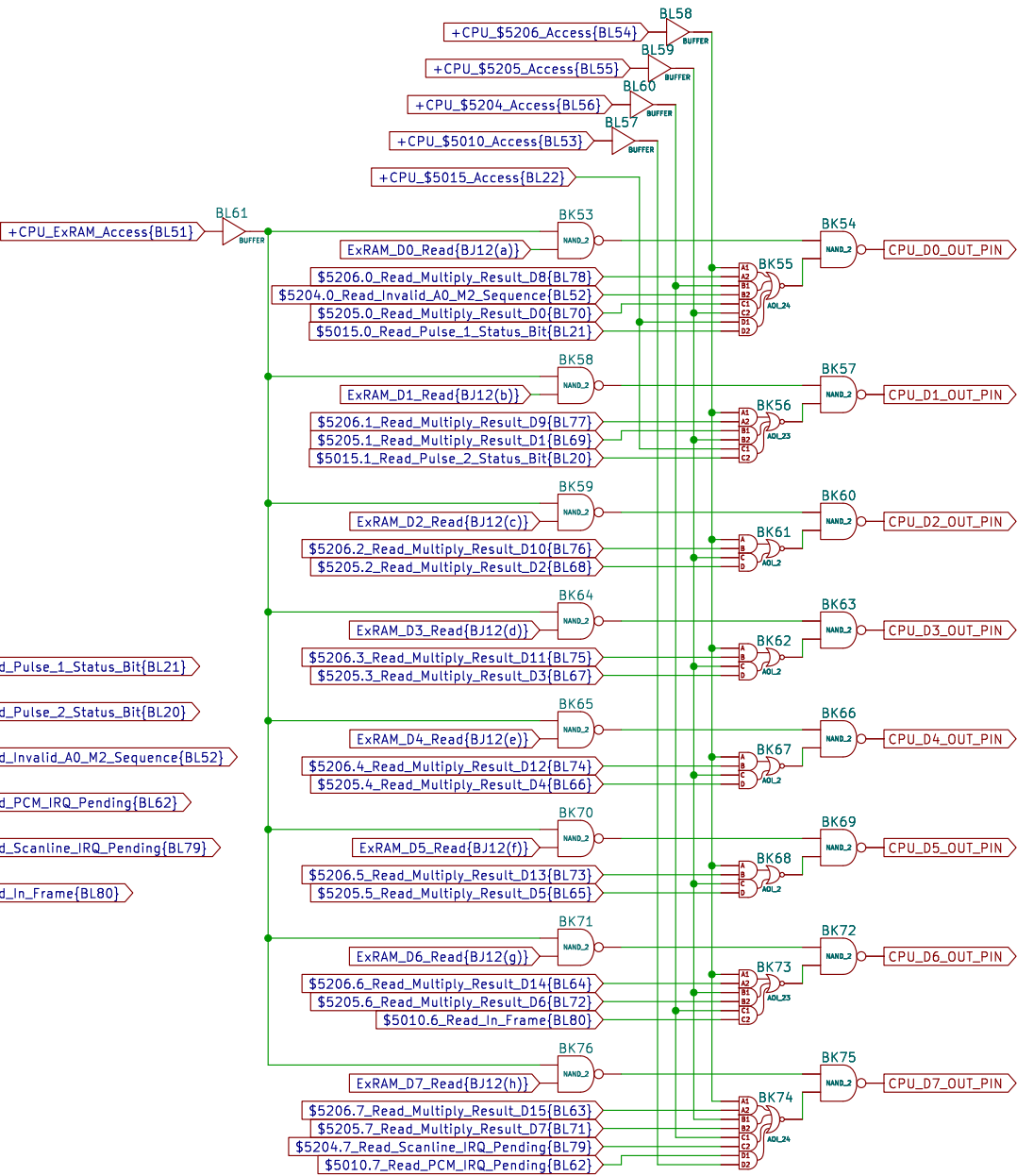
Size: USLedge | Date:
KiCad E.D.A. 9.0.7

Rev:
Id: 7/26

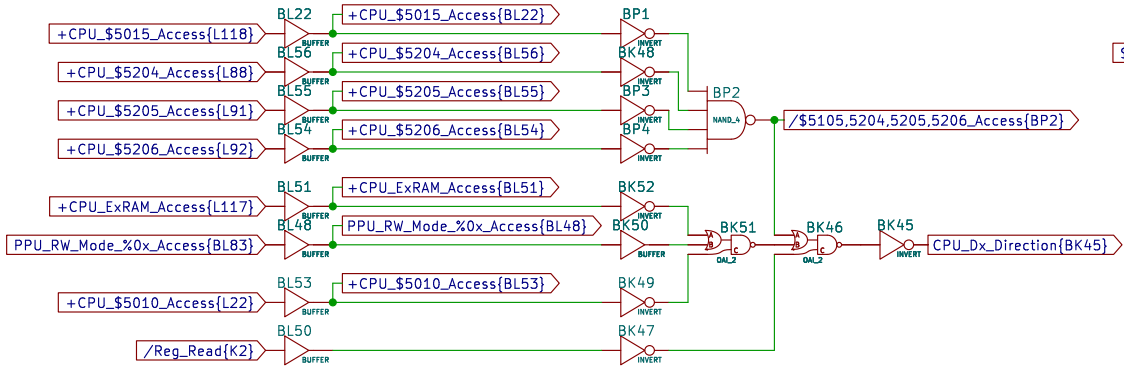
Incoming CPU Data Bus Buffering Bonanza



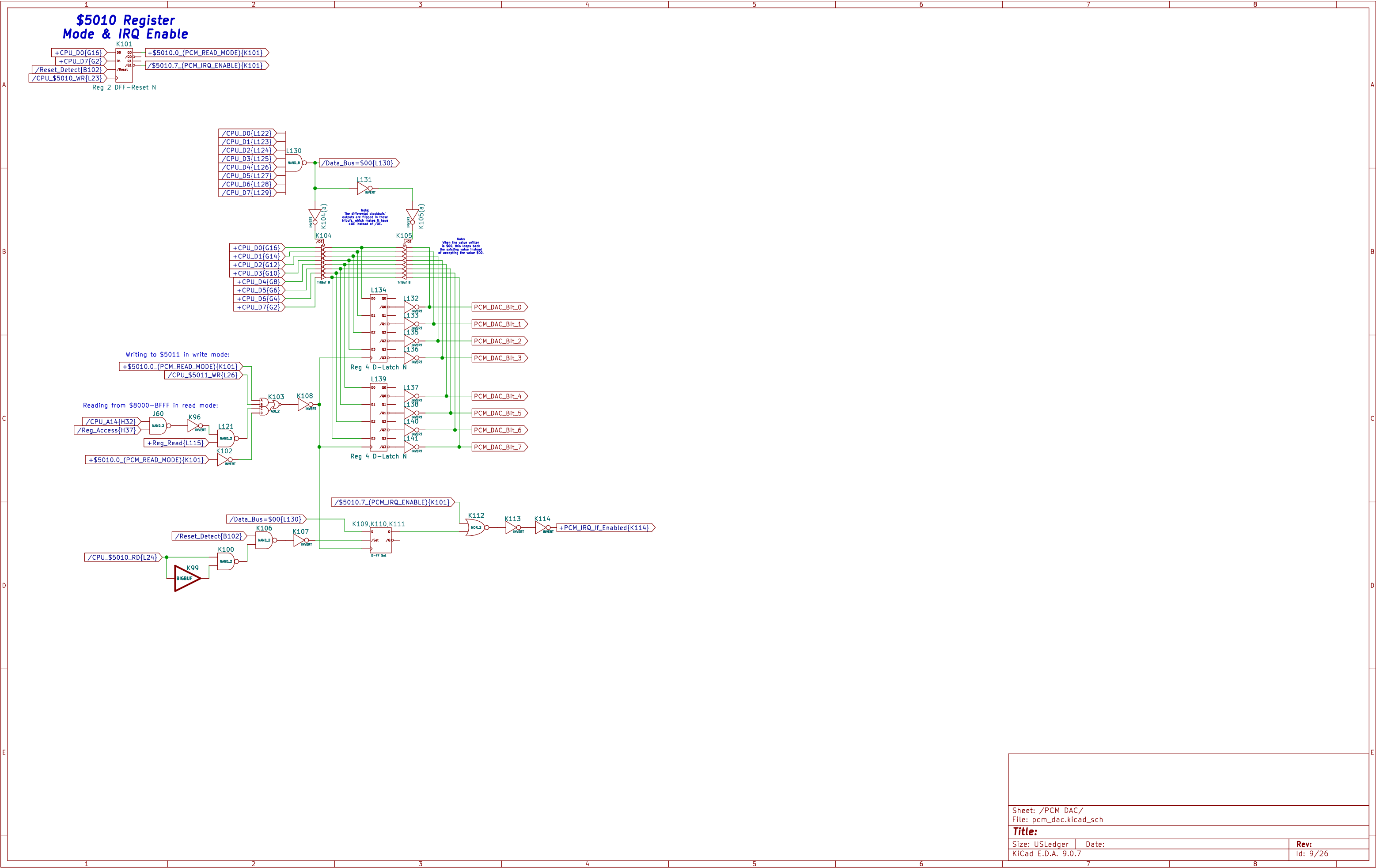
Multiplexing Data to be Driven onto the CPU Data Bus

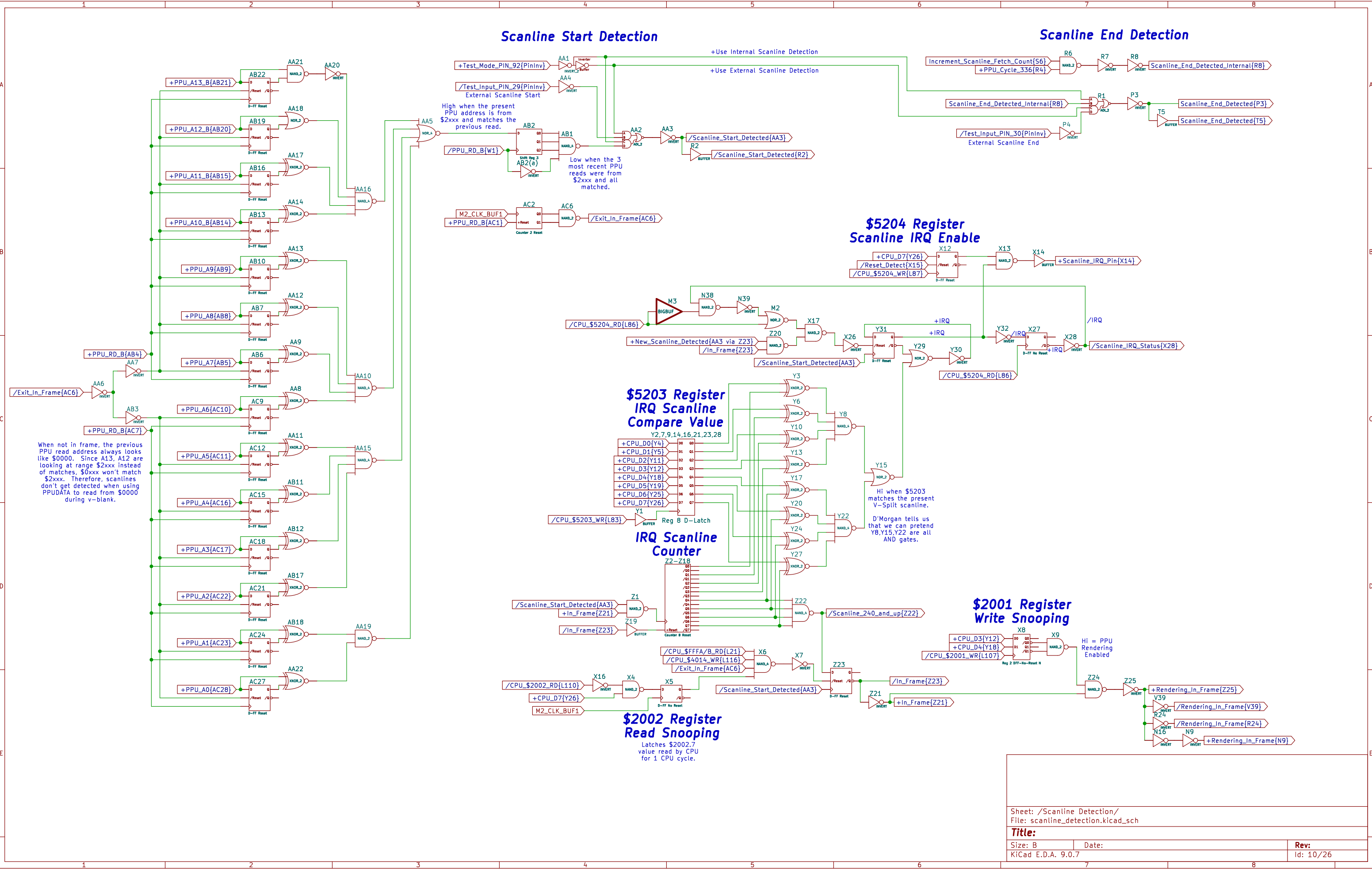


Decides When to Drive the CPU Data Bus



\$5010 Register Mode & IRQ Enable





When not in frame, the previous PPU read address always looks like \$0000. Since A13, A12 are looking at range \$2xxx instead of matches, \$0xxx won't match \$2xxx. Therefore, scanlines don't get detected when using PPU DATA to read from \$0000 during v-blank.

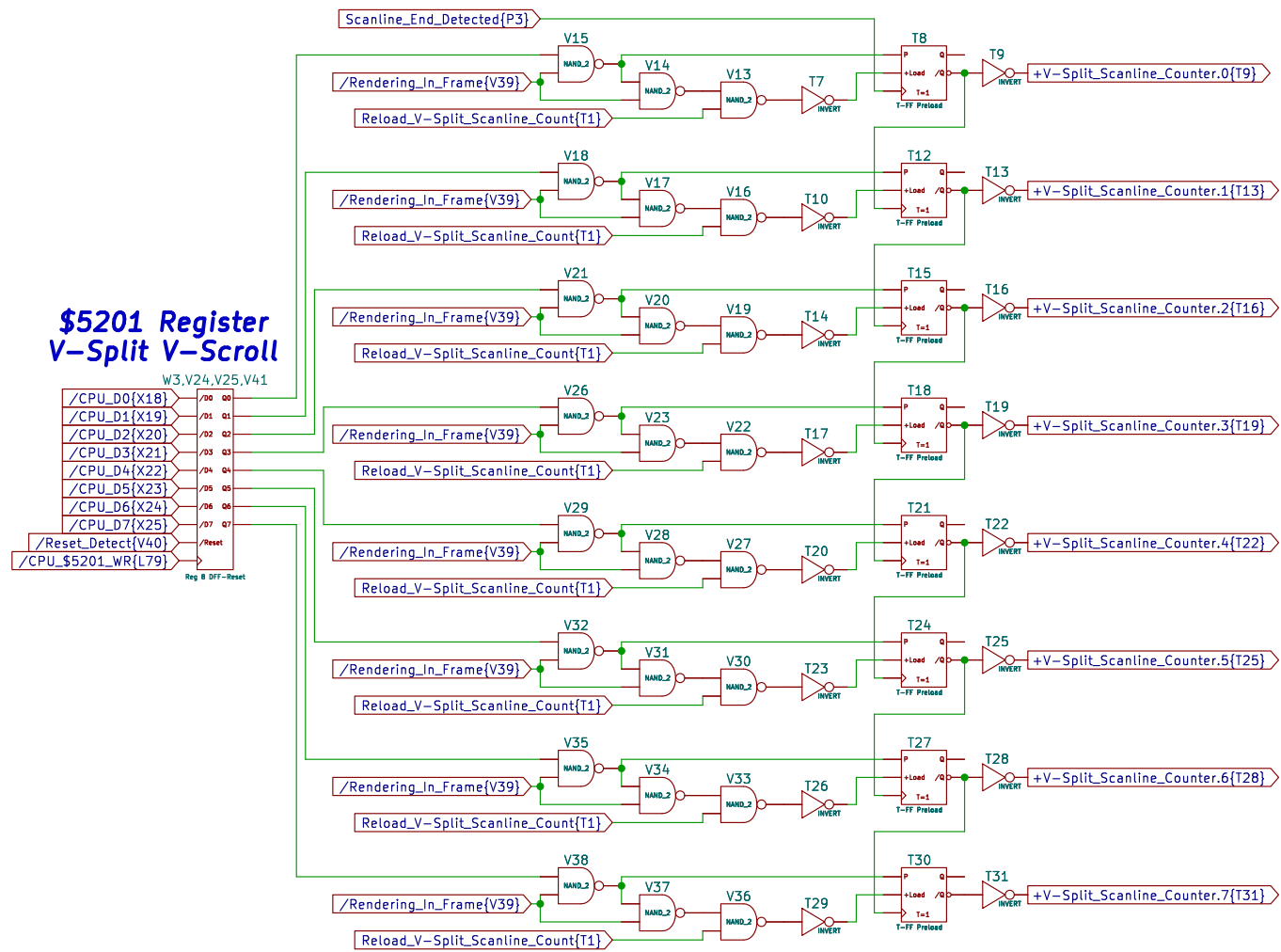
High when the present PPU address is from \$2xxx and matches the previous read.

Low when the 3 most recent PPU reads were from \$2xxx and all matched.

Hi when \$5203 matches the present V-Split scanline.

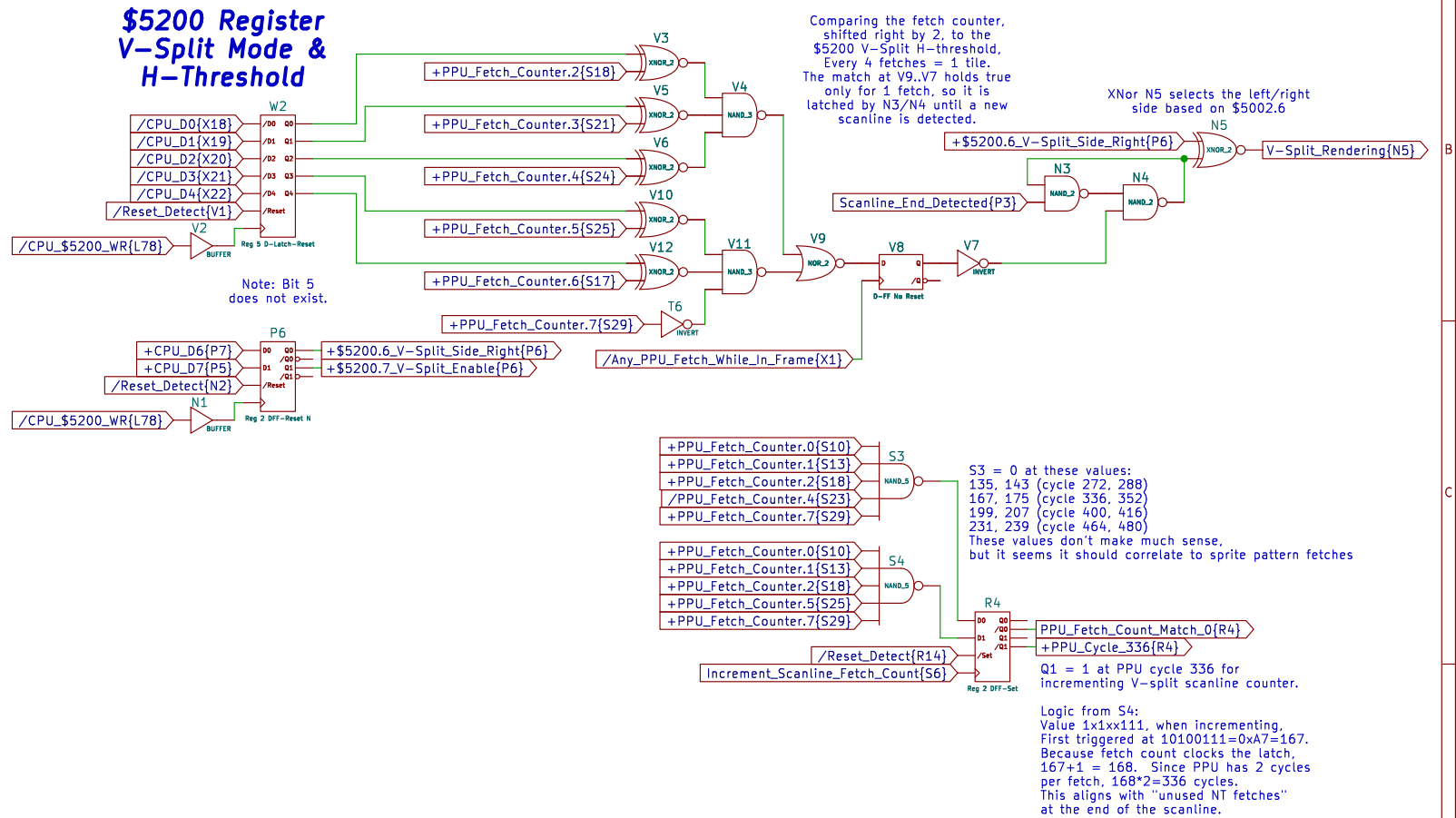
D'Morgan tells us that we can pretend Y8,Y15,Y22 are all AND gates.

V-Split Scanline Counter



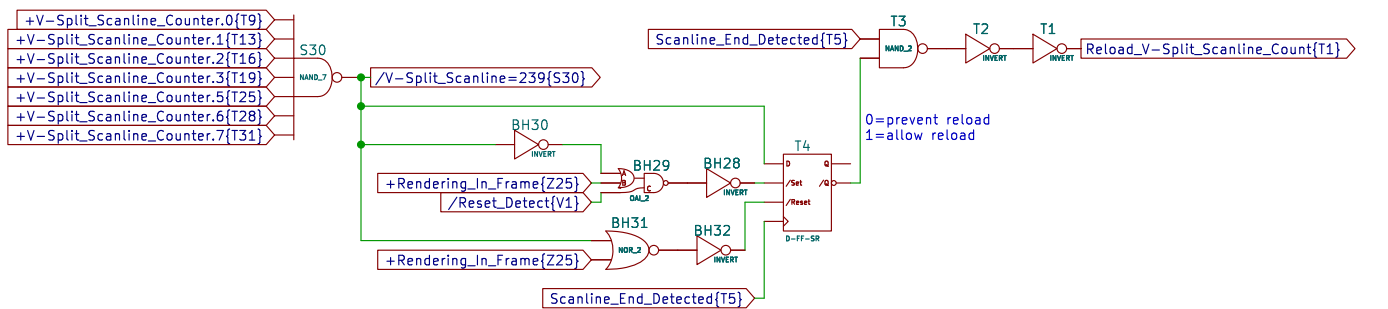
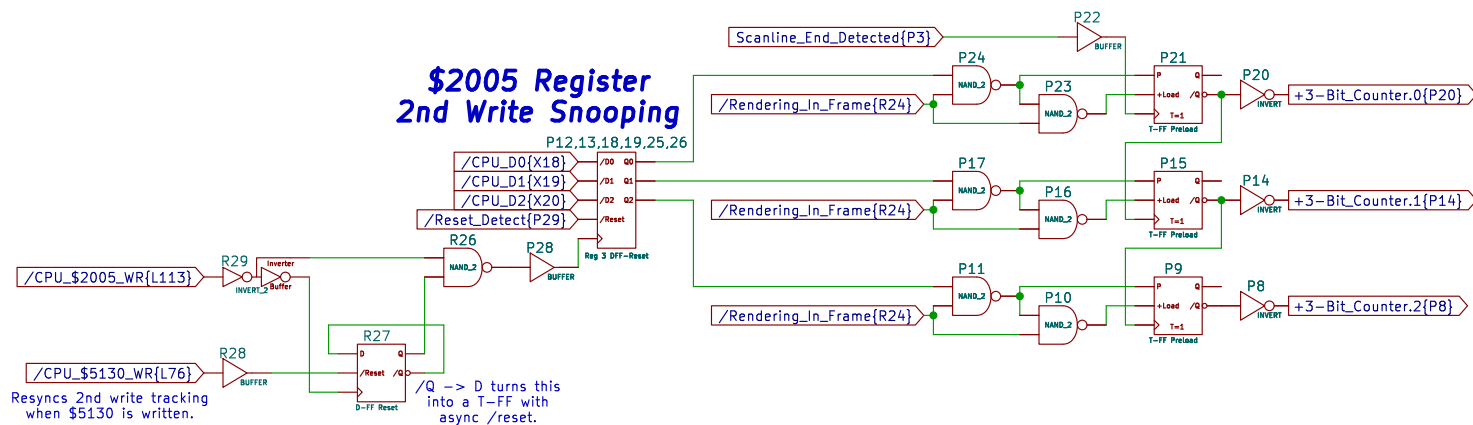
\$5201 Register V-Split V-Scroll

\$5200 Register V-Split Mode & H-Threshold



Unknown 3-Bit Scanline Counter

\$2005 Register 2nd Write Snooping



Sheet: /V-Split/
File: v_split.kicad_sch

Title:

Size: B

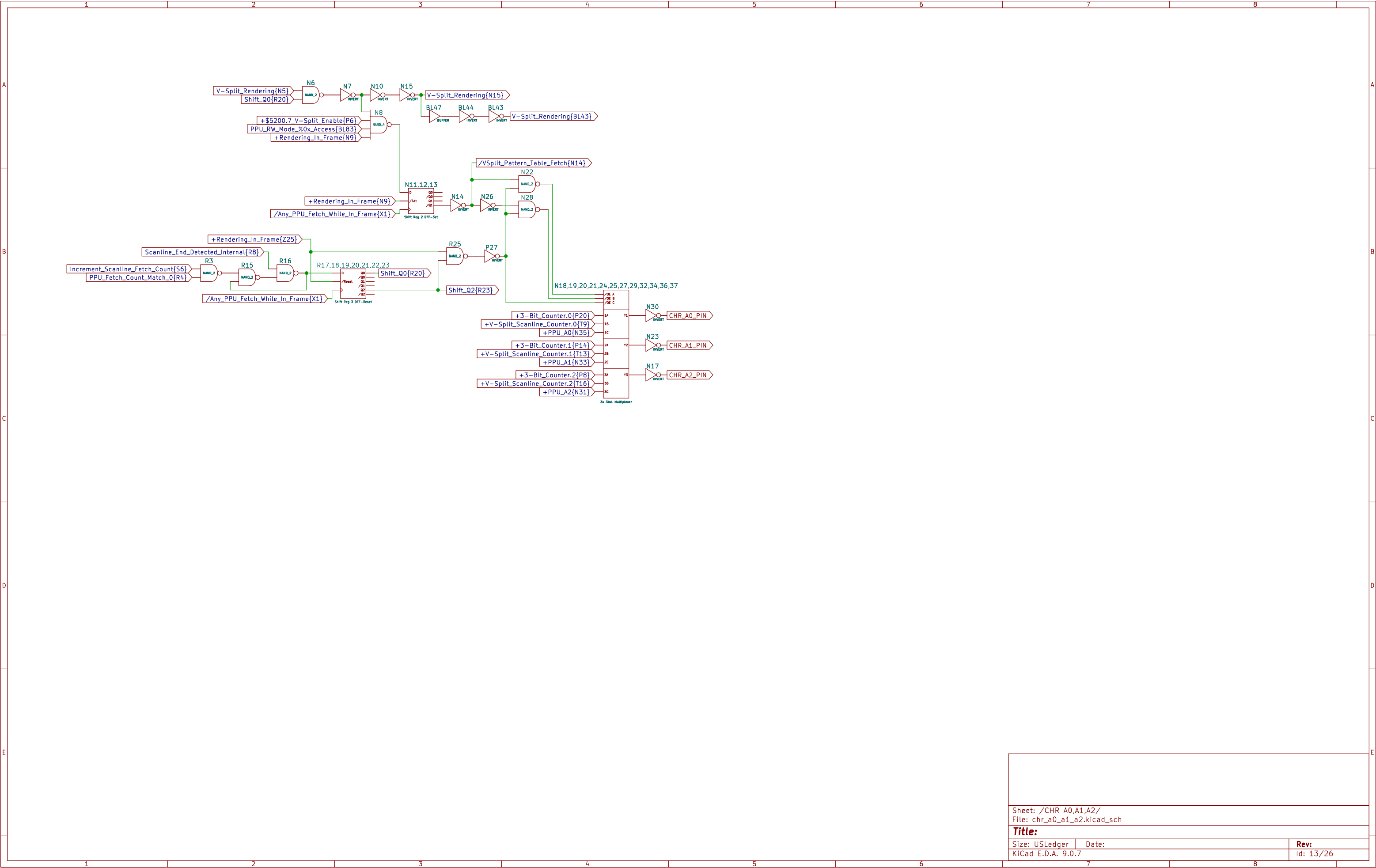
Date:

Rev:

KiCad E.D.A. 9.0.7

Id: 11/26

1	2	3	4	5	6	7	8	
---	---	---	---	---	---	---	---	--



\$5104 Register ExRAM Mode

pattern tables
ated to
g used here.

about PPU /A13
PU has control of
?

used as a nametable
is set to CPU mode.

That doesn't make sense because
what if you have a nametable
assigned to ExRAM in \$5105?
It needs ExRAM Ax coming from P

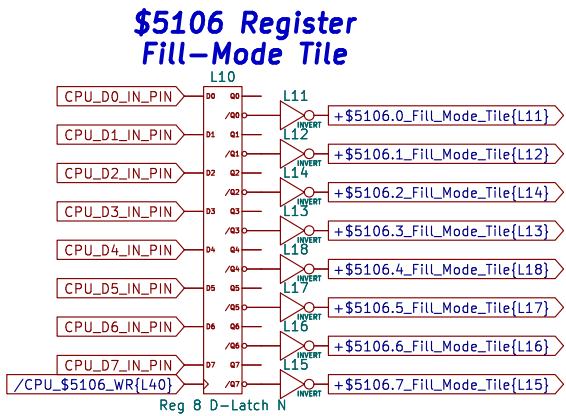
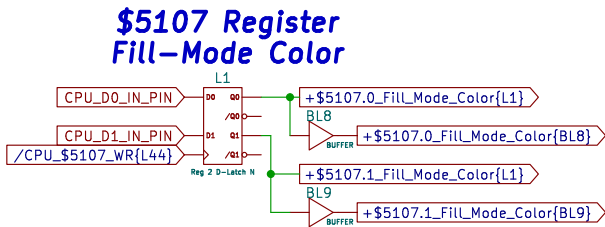
Consider:
ExRAM can still be used as a nametable
when ExRAM mode is set to CPU mode.

+PPL
+ExRAM

Title:

Size: USLedge	Date:
KiCad E.D.A. 9.0.7	

Rev: 14/26



Sheet: /Fill Mode/
File: fill_mode.kicad_sch

Title:

Size: USLedger

Date:

Rev:

KiCad E.D.A. 9.0.7

Id: 15/26

See spreadsheet for decoded values:
"pulse length lookup table.xlsx"

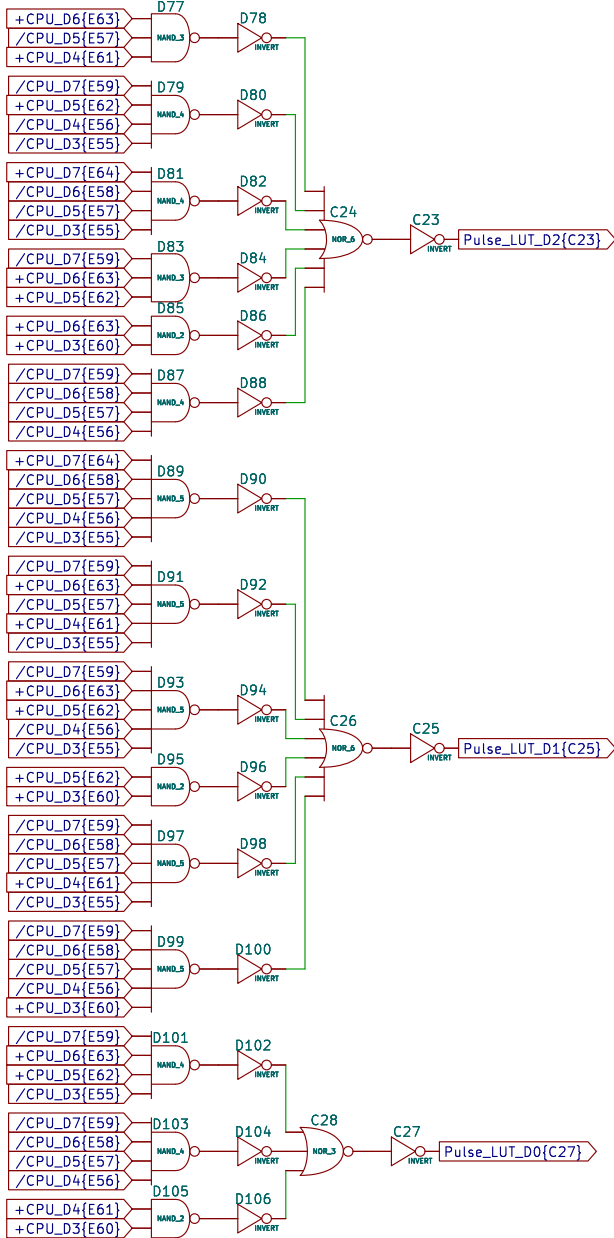
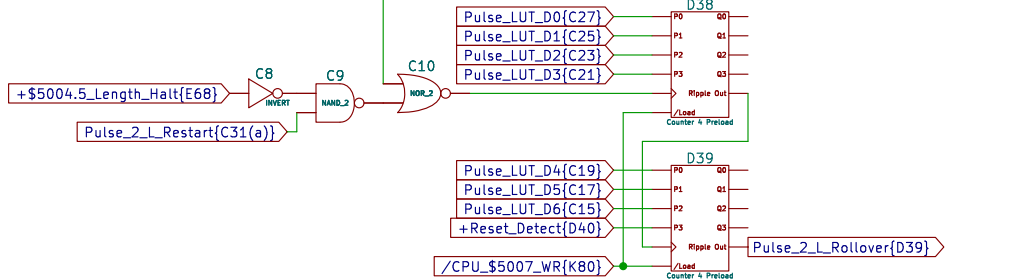
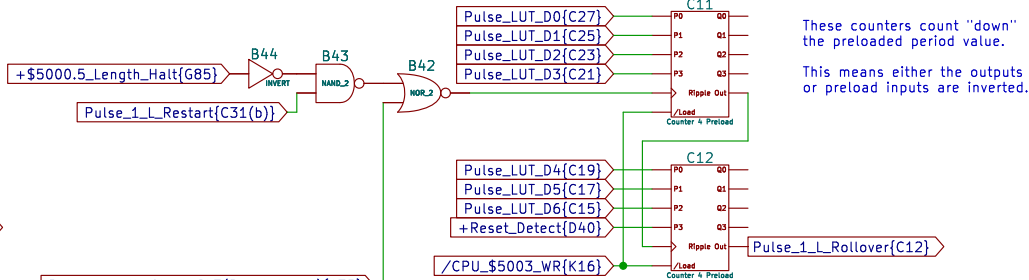


Diagram showing the connection of four LUT outputs to a 4-bit parallel adder. The LUT outputs are Pulse_LUT_D0[C27], Pulse_LUT_D1[C25], Pulse_LUT_D2[C23], and Pulse_LUT_D3[C21]. These are connected to the inputs P0, P1, P2, and P3 of the 4-bit Parallel Adder. The adder also has a Carry Out output.

[illegible]

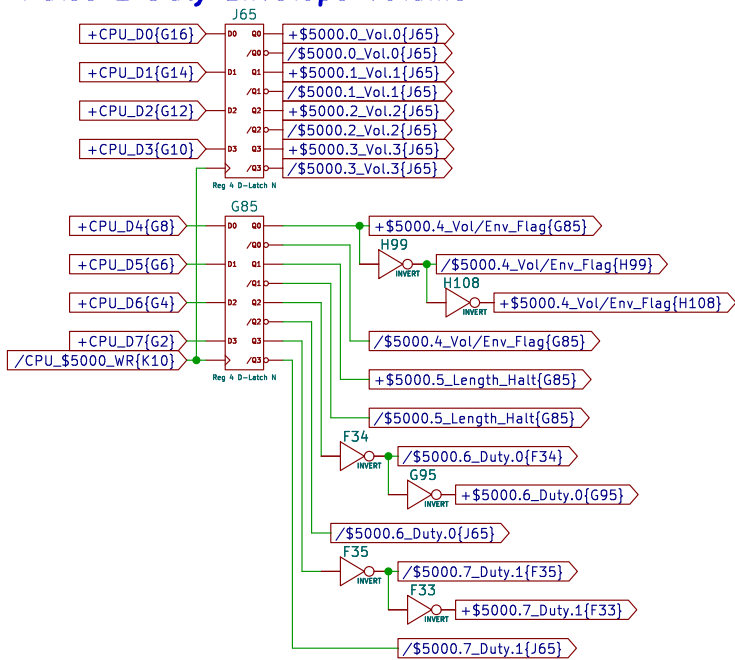
All of these outputs are
241.08 Hz
-or-
Pin 98 Input

Sheet: /Pulse Length Lookup Table/
File: pulse_length_lookup_table.kicad_sch

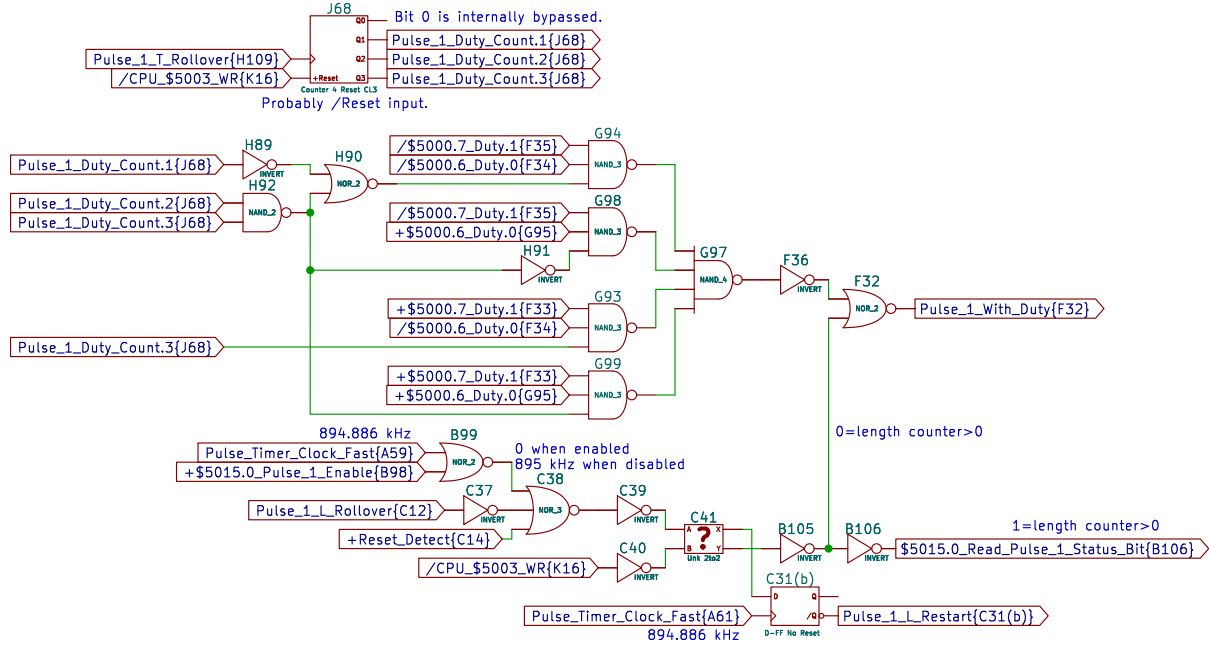
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Size: USLedger	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 16/26

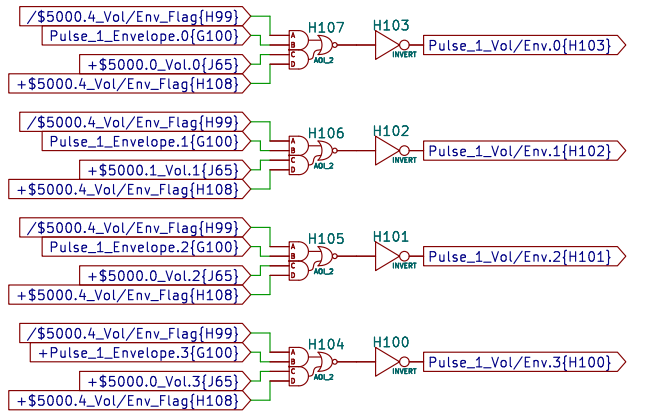
\$5000 Register Pulse 1 Duty Envelope Volume



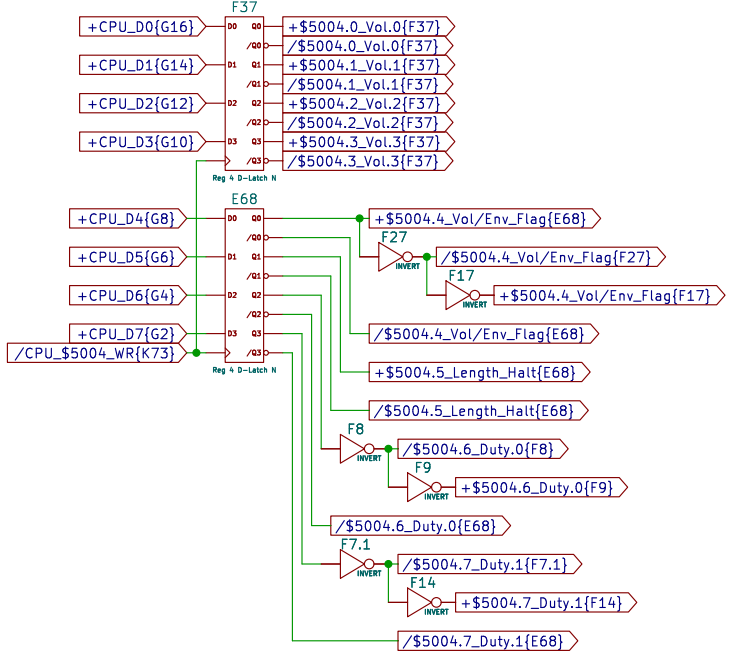
Pulse 1 Duty Cycle Counter



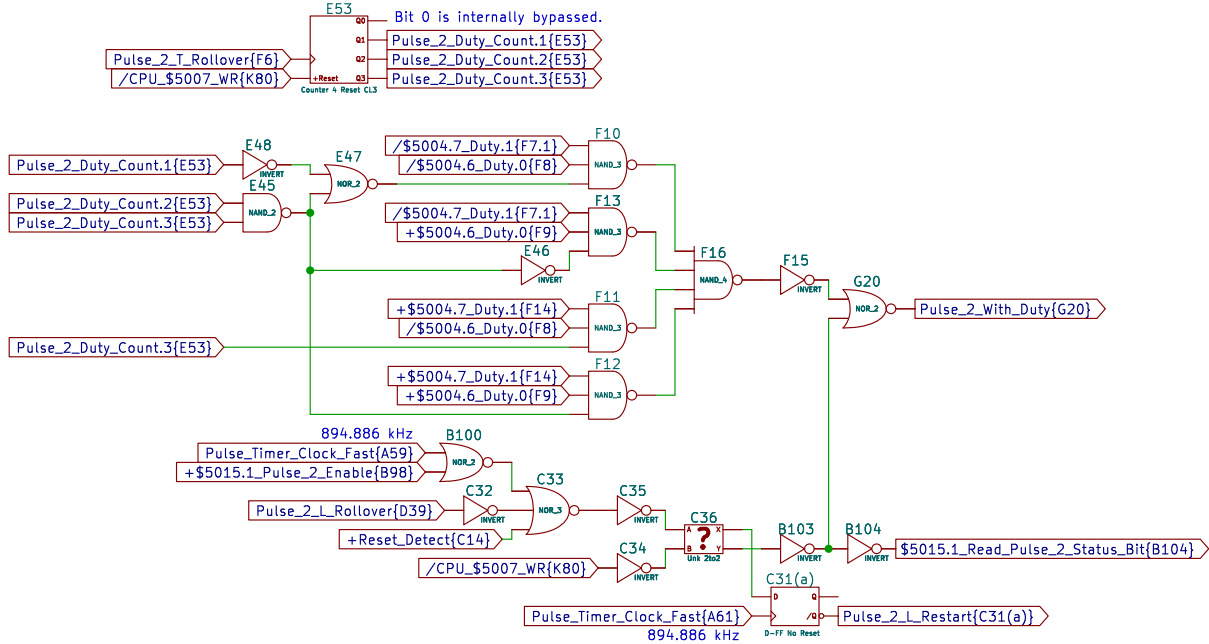
Pulse 1 Envelope/Volume Multiplexing



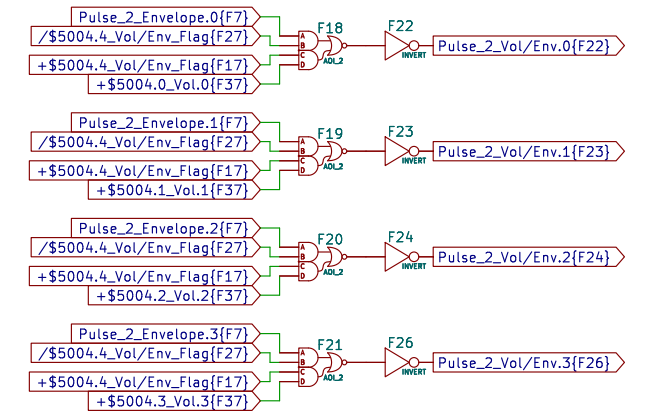
\$5004 Register Pulse 2 Duty Envelope Volume



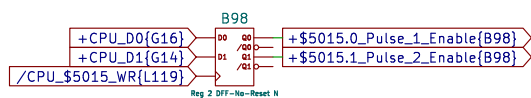
Pulse 2 Duty Cycle Counter



Pulse 2 Envelope/Volume Multiplexing



\$5015 Write Register Pulse 1&2 Status

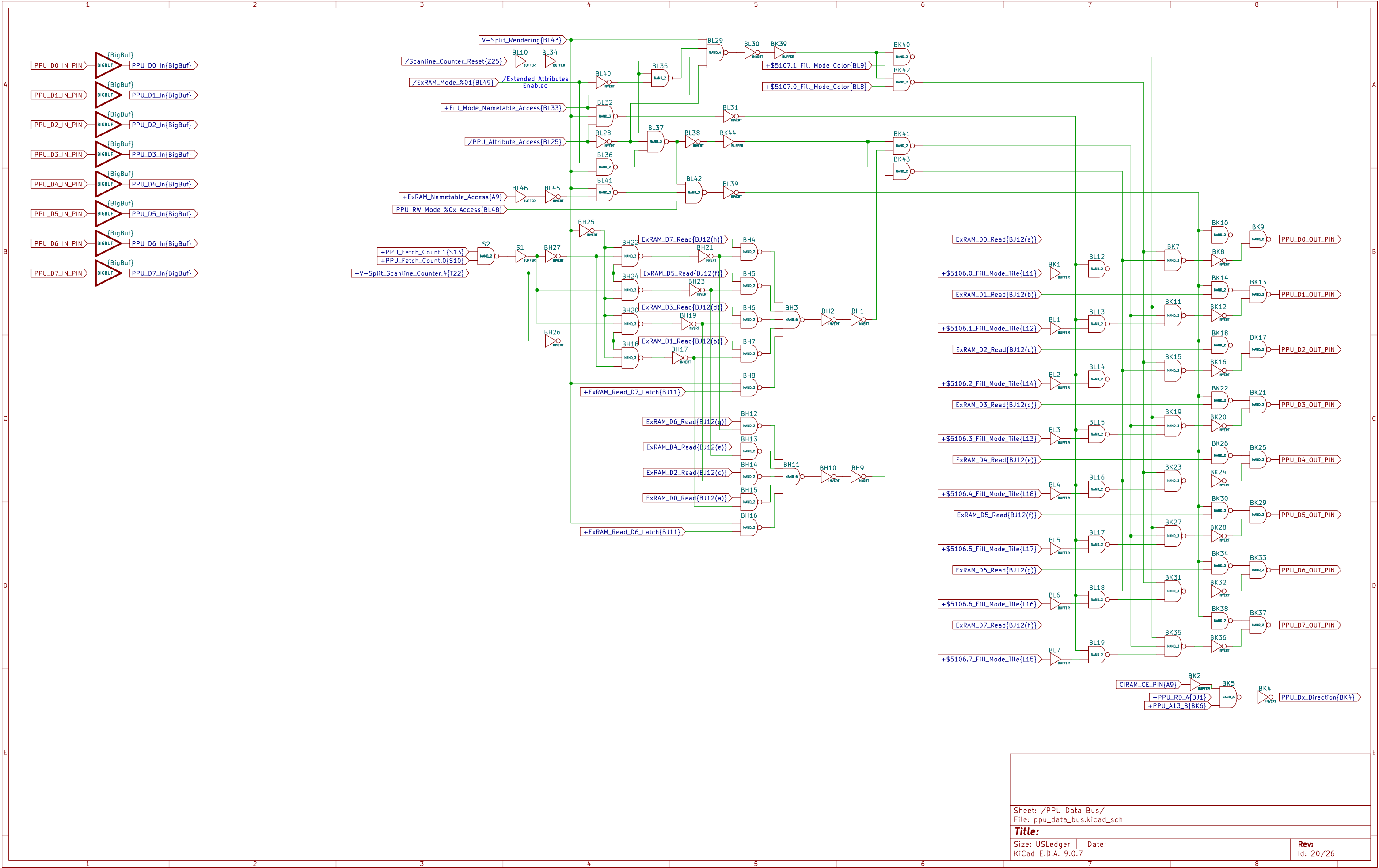


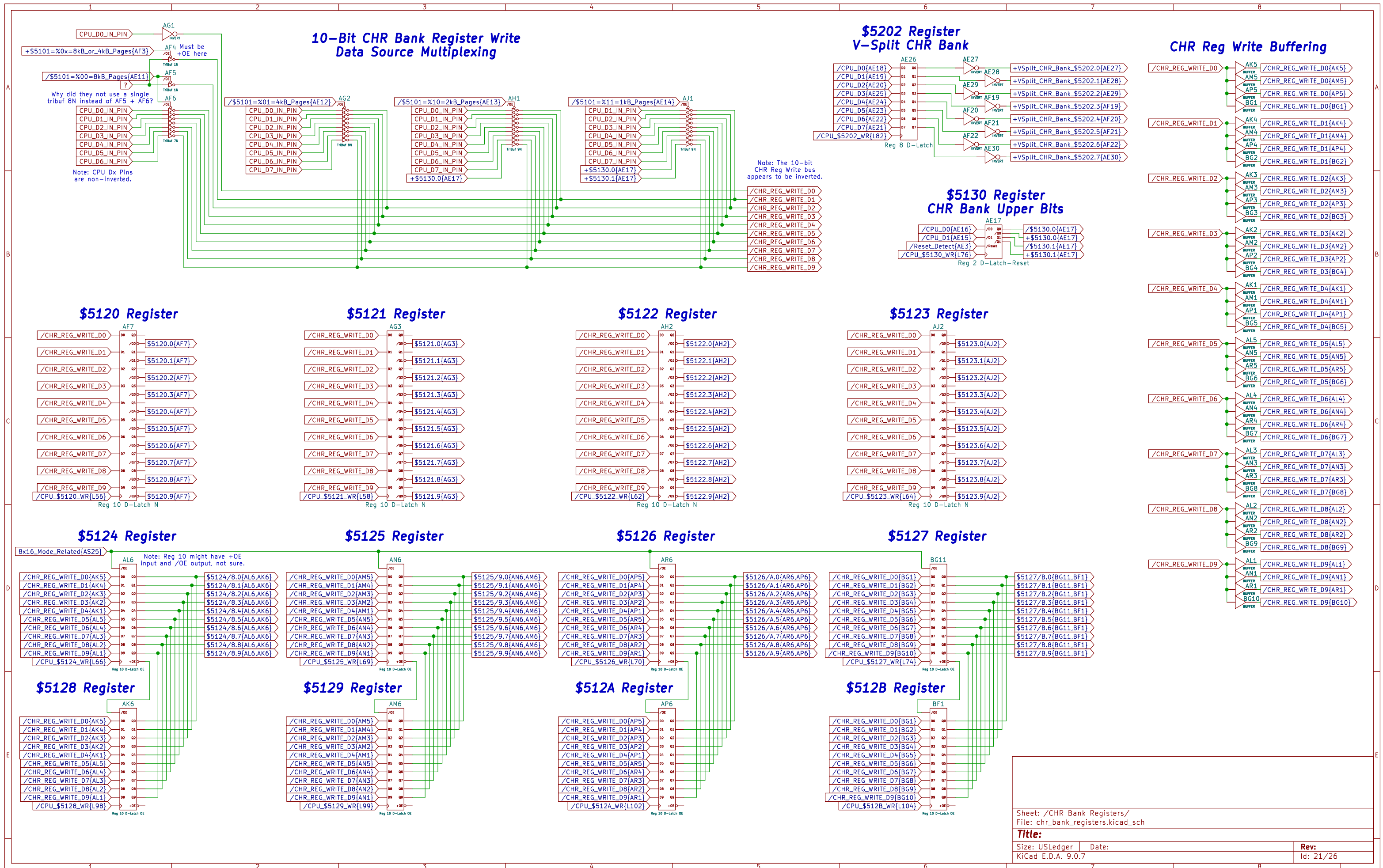
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File: pulse_duty_envelope_volume.kicad_sch

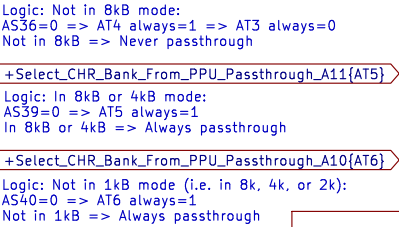
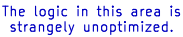
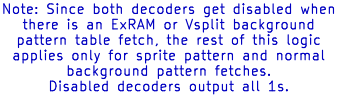
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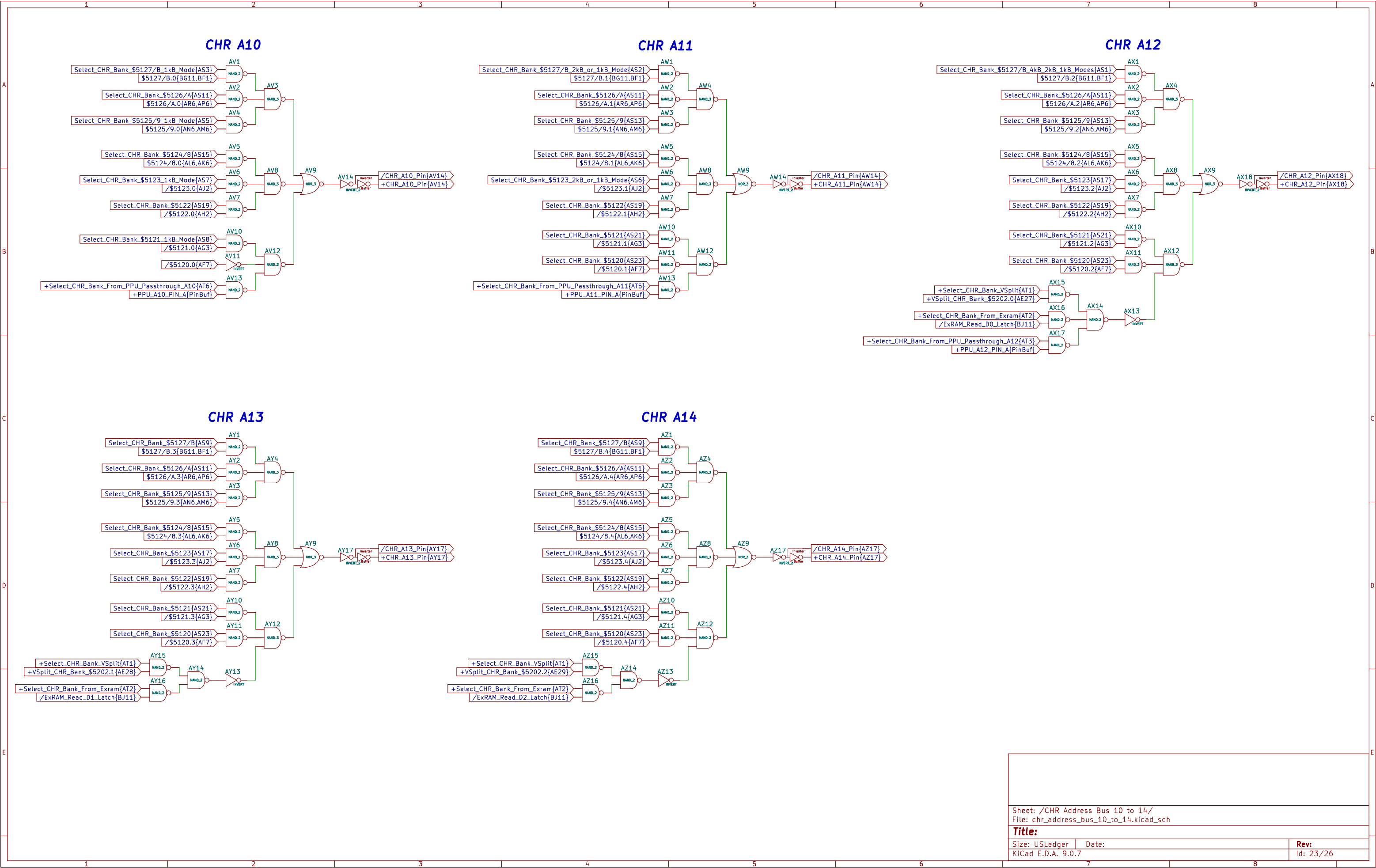
Size: USLdger Date:
KiCad E.D.A. 9.0.7

Rev:
Id: 18/26

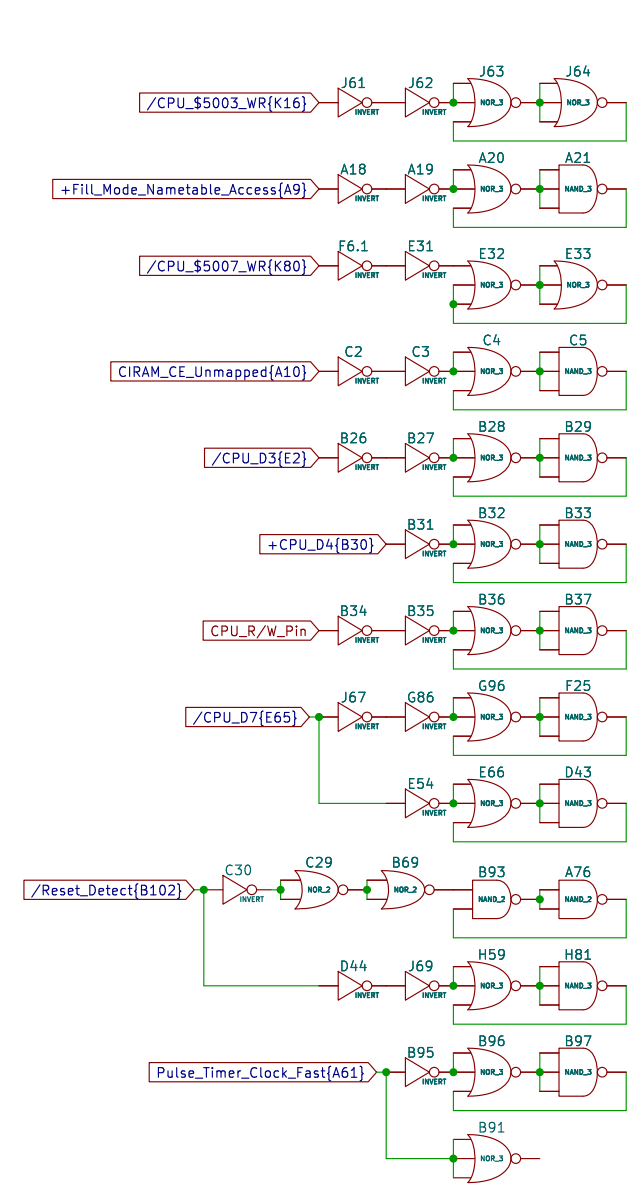




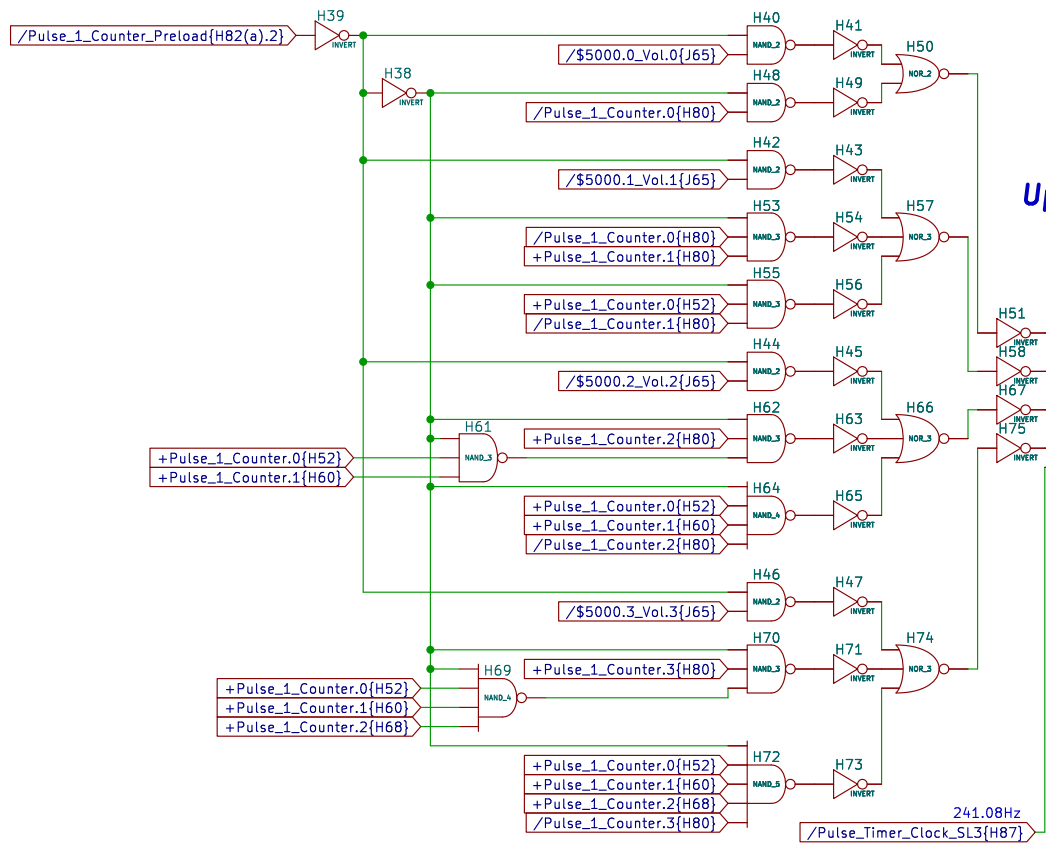
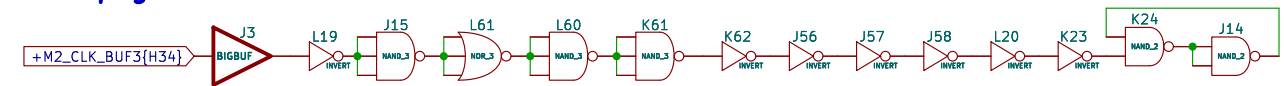




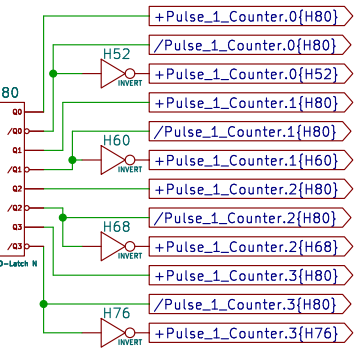
This sheet contains all residual items that don't serve any apparent purpose.



M2 Spaghetti That Goes Nowhere

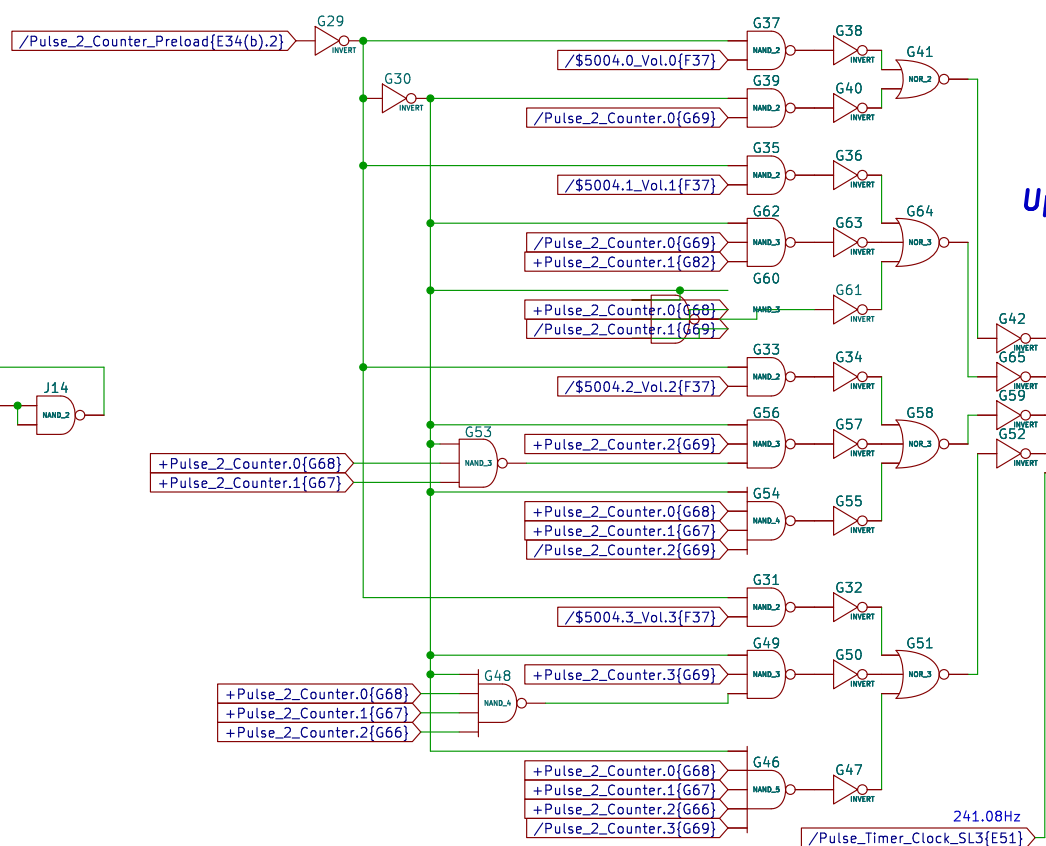


Pulse 1
Unused 4-bit Binary
Up-Counter With Preload

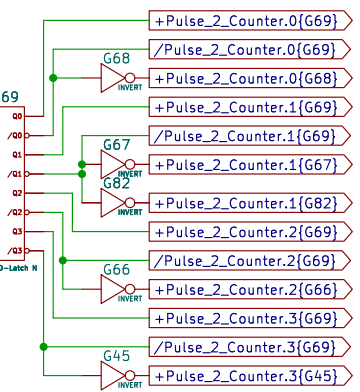


The only output that goes elsewhere is H78, which gets latched by H82(a), coming directly back here as /Pulse_1_Counter_Preload; all self-contained. It does not appear to provide any other external signals, so I don't think it actually does anything. It seems like an abandoned envelope counter that was left in.

It latches the 4-bit pulse volume, then counts up at a rate of 240 Hz. Nothing stops it from counting or rolling over from \$F back to \$0.



Pulse 2
Unused 4-bit Binary
Up-Counter With Preload



Sheet: /Artifacts/
File: artifacts.kicad_sch
Title:
Size: USLdger Date:
KiCad E.D.A. 9.0.7
Rev:
Id: 26/26